

Probabilistic Analysis and Quick Sort Notes

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Date: Oct 3, 2026

Introduction

This technical note summarizes the tools and logic used in probabilistic analysis.

Type of probabilistic analysis

There are four main types of probabilistic analysis.

- Random if
 - With Recursion
 - Without recursion
- Random for
 - Single random loop
 - Random loop inside an outer loop
- Random-sized recursive call (i.e. Number line problem)
- Loop that randomly terminates

1. Random if

Method: *Find the probabilities of both conditions and their corresponding $T(n)$, then use the formula.*

1.1 Random if with recursion

```
1. k = Random(n)
2. if Boolean condition on k then
3.     Make recursive call
4. else
5.     Make recursive call
6. end if
```

For example

```
function Function1(A[], n) {
    k = random(n)
    if (k mod 4 = 1) {
        x = x + Function1(A[], n/2)
    } else {
        x = x + Function1(A[], n/3)
    }
}
```

We can use the following formula:

$$ET(n) = \Pr(A) * E(T(n) | A) + \Pr(\text{Not } A) * E(T(n) | \text{Not } A)$$

OR

$$ET(n) = \Pr(A) * E(T(n) | A) + \Pr(\text{Not } A) * E(T(n) | \neg A)$$

From the formula, we can easily see that

$$ET(n) = \frac{1}{4} * T\left(\frac{n}{2}\right) + \left(1 - \frac{1}{4}\right) * T\left(\frac{n}{3}\right)$$

1.2 Random if without recursion

```
1. k = Random(n)
2. if Boolean condition on k then
3.     Foo()
4. else
5.     Bar()
6. end if
```

For example

```
function Function2(A[], n) {
    k = random(n)
    int x = 0;
    if (k mod 4 = 1) {
        for (i = 1 to n) {
            x = x + A[i];
        }
    } else {
        int i = 1;
        while (i <= n) {
            x = x + A[i];
            i = 2 * i;
        }
    }
}
```

We can use the following formula (Same as in Section 1.1):

$$ET(n) = \Pr(A) * E(T(n) \mid A) + \Pr(\text{Not } A) * E(T(n) \mid \neg A)$$

OR

$$ET(n) = \Pr(\text{Foo}) * E(T(n) \mid \text{Foo}) + \Pr(\text{Bar}) * E(T(n) \mid \text{Bar})$$

Since

$$\Pr(A) = \frac{1}{4}, E(T(n) \mid \text{Foo}) = \Theta(n)$$

and

$$\Pr(\neg A) = \frac{3}{4}, E(T(n) \mid \text{Bar}) = \Theta(\log n)$$

From the formula, we can easily get that

$$ET(n) = \frac{1}{4} * \Theta(n) + \frac{3}{4} * \Theta(\log n)$$

2. Random for

Method: *First analyze the running time for a fixed random value of k, and then incorporate the result into the outer loop if there is one.*

2.1 single random loop

1. k = Random(n)
2. Loop that runs f(k) times

For example

```
function Function3(A[], n) {
    x = 0;
    k = Random(n);
    for (i = 1 to k){
        for(j = 1 to i * i) {
            x = x + A[ij mod n];
        }
    }

    return x;
}
```

We can use the formula:

$$ET(n) = \sum_{q=1}^n \Pr(k = q) t(k = q)$$

We should first find $t(k=q)$, which is the running time of the loop controlled by k .

$$t(k = q) = \left(\sum_{i=1}^q i^2 C \right)$$

Since k is uniformly random,

$$\Pr(k = q) = \frac{1}{n}$$

Therefore,

$$ET(n) = \sum_{q=1}^n \frac{1}{n} \sum_{i=1}^q i^2 C$$

2.2 Random loop inside an outer loop

```
1. loop
2.     k = Random(i)
3.     Loop that runs f(i) times
4. end loop
```

For example

```
function Function4(A[], n) {
    x = 0;
    for (i = 1 to n) {
        k = Random(i);
        for (j = 1 to k * k){
            x = x + A[(i*j mod n)]
        }
    }

    return x;
}
```

For each iteration of the outer loop, we first analyze the time complexity of the inner loop. (Same as what we did in Section 2.1)

We can use the formula:

$$ET_i(n) = \sum_{q=1}^i \Pr(k = q) t(k = q)$$

Since k is uniformly random from 1 to i ,

$$\Pr(k = q) = \frac{1}{i}$$

and because the inner loop runs k^2 times,

$$t(k = q) = q^2 C$$

Therefore,

$$\begin{aligned} ET_i(n) &= \sum_{q=1}^i \frac{1}{i} \Theta(q^2) \\ &= \frac{1}{i} \Theta(i^3) = \Theta(i^2) \end{aligned}$$

Then, we can determine the overall time complexity.

$$\begin{aligned} ET(n) &= \sum_{i=1}^n ET_i(n) \\ &= \sum_{i=1}^n \Theta(i^2) = \Theta(n^3) \end{aligned}$$

3. Random-sized recursive call (i.e. Number Line problem)

Method: *First identify the best and worst cases. Then use a number line to divide the possible values of k into ranges, and use the appropriate worst-case bound in each range to determine the overall time complexity.*

```
1. Base case and non-recursive work
2. Do ( )
3.     k = Random(n)
4.     Make recursive call of size f(k).
4. return
```

For example

```

function Function5(A[], n) {
    if (n <= 22) {
        return 1;
    }
    x = 0;
    for (i = 1 to n/2) {
        for (j = 1 to n/4) {
            x = x + A[(i*j mod n)];
        }
    }
    k = Random(n/5 - 1);
    x = x + Func5(A[], 5k);
    x = x + Func5(A[], n - 5k);

    return x;
}

```

We use the following steps:

1. Substitute the minimum and maximum possible values of k and compare the resulting recursive-call sizes to identify the best and worst cases.
2. If the minimum and maximum values of k produce the same case, also test the midpoint

$$k = \frac{k_{\min} + k_{\max}}{2}$$

to determine whether it gives the best case.

3. Divide the possible values of k into several ranges using the critical points on the number line.
4. For each range, choose the value of k that gives the worst recursive-call sizes, then use these cases to construct the upper bound.
5. For the lower bound, use the best-case recursive-call sizes.

Solution:

At the two endpoints,

$$k = 1 \Rightarrow T(5) + T(n - 5)$$

and

$$k = \frac{n}{5} - 1 \Rightarrow T(n - 5) + T(5).$$

Both endpoints give the same unbalanced recursive-call sizes.

To find the midpoint, set the two recursive-call sizes equal:

$$5k = n - 5k$$

$$10k = n$$

$$k = \frac{n}{10}.$$

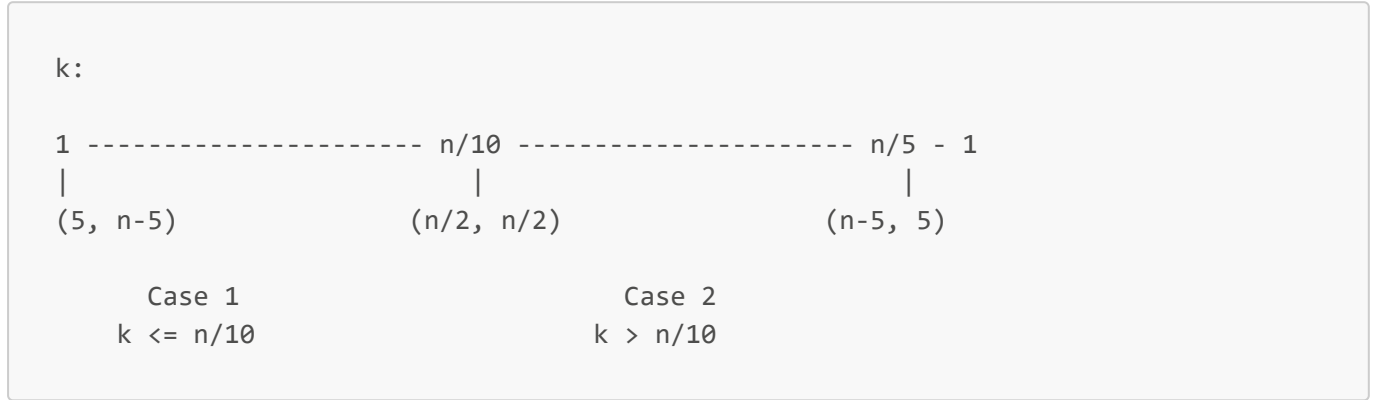
Therefore,

$$5k = n - 5k = \frac{n}{2},$$

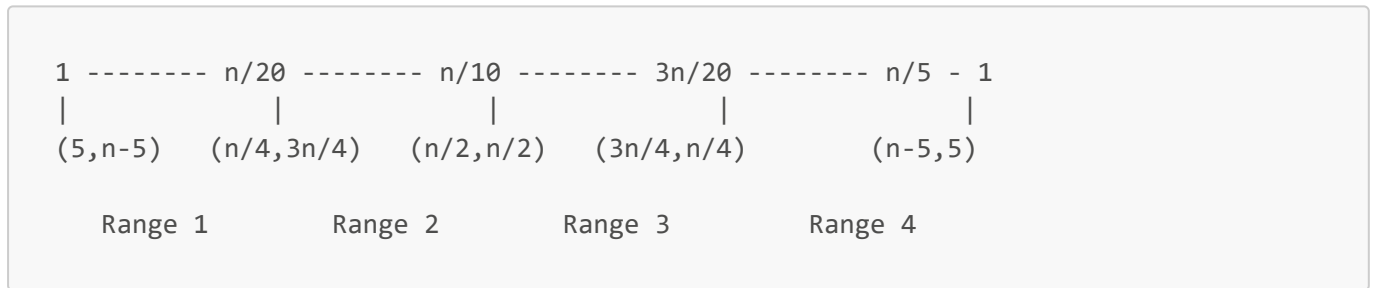
so the recursive cost at the midpoint is

$$2T(n/2).$$

Then we draw the number line:



Next, divide the possible values of k into four ranges:



For the upper bound, choose the worst recursive-call sizes in each range.

Therefore,

$$ET(n) \leq C_1 n^2 + \frac{1}{4} [ET(5) + ET(n-5)] + \frac{1}{4} [ET(n/4) + ET(3n/4)]$$

$$+ \frac{1}{4} [ET(3n/4) + ET(n/4)] + \frac{1}{4} [ET(n-5) + ET(5)].$$

Combining identical terms gives

$$ET(n) \leq C_1 n^2 + \frac{1}{2} [ET(5) + ET(n-5)] + \frac{1}{2} [ET(n/4) + ET(3n/4)].$$

Since $ET(5) = \Theta(1)$, it can be absorbed into the other asymptotic terms.

For the upper bound, we can also replace $ET(n-5)$ by $ET(n)$ because

$$n-5 < n$$

and therefore

$$ET(n-5) \leq ET(n).$$

Thus,

$$ET(n) \leq C_2 n^2 + \frac{1}{2} ET(n) + \frac{1}{2} ET(n/4) + \frac{1}{2} ET(3n/4)$$

Move $\frac{1}{2} ET(n)$ to the left-hand side:

$$ET(n) - \frac{1}{2} ET(n) \leq C_2 n^2 + \frac{1}{2} ET(n/4) + \frac{1}{2} ET(3n/4)$$

Therefore,

$$\frac{1}{2} ET(n) \leq C_2 n^2 + \frac{1}{2} ET(n/4) + \frac{1}{2} ET(3n/4)$$

Multiplying both sides by 2 gives

$$ET(n) \leq ET(n/4) + ET(3n/4) + Cn^2$$

Therefore, using a recursion tree, we obtain

$$ET(n) = O(n^2).$$

For the Lower Bound, the expected running time cannot be better than the best possible recursive split.

The best split occurs at

$$k = \frac{n}{10},$$

which gives two recursive calls of size $n/2$.

Therefore, the best-case recurrence is

$$T(n) = 2T(n/2) + Cn^2.$$

This recurrence gives

$$T(n) = \Theta(n^2).$$

Since the expected running time cannot be better than the best-case running time,

$$ET(n) = \Omega(n^2).$$

Therefore,

$$\boxed{ET(n) = \Theta(n^2)}$$

4. Loop that randomly terminates

1. Do some work
2. repeat
3. $k = \text{Random}(n)$
4. Do some work
5. until Boolean condition on k is true

For this type of problem, the loop terminates only when a random condition is satisfied.

We let X be the number of times the loop repeats before termination.

For example:

```
function Function6(A[], n) {
    x = 0;

    repeat {
        k = Random(n);

        for (i = 1 to sqrt(n)) {
            x = x + A[i];
        }

    } until (k == 42);

    return x;
}
```

The loop terminates when

$$k = 42.$$

Since k is uniformly random from 1 to n ,

$$\Pr(k = 42) = \frac{1}{n}$$

and

$$\Pr(k \neq 42) = \frac{n-1}{n}.$$

For the expected case, let X be the number of times the loop repeats before termination.

The running time can be written as

$$T(n) = c\sqrt{n} + X \cdot c\sqrt{n}.$$

Therefore,

$$ET(n) = c\sqrt{n} + E(X) \cdot c\sqrt{n}.$$

To find $E(X)$, use

$$E(X) = \sum_{i=0}^{\infty} i \Pr(X = i).$$

Then,

$$E(X) = \sum_{i=1}^{\infty} \Pr(X \geq i).$$

For the loop to repeat at least i times, the terminating condition must fail repeatedly. Thus,

$$\Pr(X \geq i) = \left(\frac{n-1}{n}\right)^i.$$

Therefore,

$$E(X) = \sum_{i=1}^{\infty} \left(\frac{n-1}{n}\right)^i.$$

Using the geometric-series formula,

$$\sum_{i=1}^{\infty} r^i = \frac{r}{1-r},$$

we get

$$E(X) = \frac{\frac{n-1}{n}}{1 - \frac{n-1}{n}}.$$

Since

$$1 - \frac{n-1}{n} = \frac{1}{n},$$

we have

$$E(X) = \frac{n-1}{n} \cdot n = n-1.$$

Thus,

$$ET(n) = c\sqrt{n} + (n-1)c\sqrt{n}.$$

Therefore,

$$ET(n) = cn\sqrt{n}.$$

So,

$$ET(n) = \Theta(n^{3/2})$$

Reference List

1. "CSE2331_Types_of_Probabilistic_Problems" Created by Professor Painter.
2. "Quick Sort Worksheet AU26" Created by Professor Painter.
3. "CSE2331_QuickSort_and_Probabilistic_Analysis_Homework" Created by Professor Painter.